Precision = $\frac{\text{true positives}}{\text{true positives + false positives}}$ **Recall** = $\frac{\text{true positive}}{\text{true pos. + false negatives}}$

actual \ predicted	+	−
+	a	b
−	c	d

$F1 = \frac{2 \times prec \times recall}{prec. + recall}$ $sensitivity = \frac{TP}{P}$ $specificity = \frac{TN}{N}$

TP recognition rate TN recog. rate

Information gain

age	p_i	n_i	$I(p_i, n_i)$
<=30	2	3	0.971
30..40	4	0	0
>40	3	2	0.971

$Info(D) = I(9,5) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.940$

$Info_{age}(D) = \frac{5}{14}I(2,3) + \frac{4}{14}I(4,0) + \frac{5}{14}I(3,2) = 0.694$

$Gain_{age} = 0.940 - 0.694 = 0.246$

Gini Index

$gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$ $D_1 = \{low, medium\}$ 10 ins.

$D_2 =$ 4 ins.

$gini_{income} = \frac{10}{14}\left(1 - \left(\frac{7}{10}\right)^2 - \left(\frac{3}{10}\right)^2\right) + \frac{4}{14}\left(1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{6}\right)^2\right)$

Naïve Bayes

$P(buys = yes) = 9/14$ $P(buys = no) = 5/14$

- Compute $P(X|C_i)$ for each class

$P(age <= 30, buys = yes) = 2/9$ $P(student = yes, buys = no) = 1/5$

$X = (age <= 30, income = medium, student = yes)$

$P(X | buys = yes) = 0.222 \times 0.444 \times 0.667$

$P(X | buys = no) = 0.6 \times 0.4 \times 0.2$

$P(X | buys = yes) \cdot P(buys = yes) = 0.028$

$P(X | buys = no) \cdot P(buys = no) = 0.007$

Distance Between Clusters

single link: smallest distance between an elm. and an elm. in other cluster.

complete link: largest distance between an elm in one cluster and an elm in other cluster.

centroid: mean point of a cluster

medoid: most centrally located object of a cluste

$$F1_{micro} = \frac{TP}{TP + \frac{1}{2}\cdot(FP+FN)} \qquad F1_{macro} = \frac{\text{sum (F1 scores)}}{\text{num. of classes}}$$

Variance: $\sigma^2 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{N}$ t-test: $t = \frac{\overline{err}(M_1)-\overline{err}(M_2)}{\sqrt{var(M_1-M_2)/k}}$ where (k-1 degrees of freedom

$$var(M_1-M_2) = \frac{1}{k}\sum_{i=1}^{k}[err(M_1)_i - err(M_2)_i - (\overline{err}(M_1)-\overline{err}(M_2))]^2$$